

Andrey Karmanov

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Education

University of Waterloo, Wilfrid Laurier University, Computer Science & Business Admin. 09/2021 – 08/2026
Functional Programs (Racket, Haskell), OOP, Data Structures, Algorithms, Compilers, OS, Concurrent and Parallel Programming (All C++) Numerical Computation, Machine Learning (Python)

Skills

Languages: Python (Pandas, NumPy, Selenium, FastAPI), JavaScript, Typescript (React, Next.js), Ruby (Rails), C++, C, SQL
Tools: Google Cloud (S3, Cloud Run, PubSub, SQL, NoSQL), Linux, Docker, Supabase, Retool **Other:** Networking, CI/CD, HPC

Experience

Shopify, Full Stack Software Engineer 01/2025 – 05/2025
- Working with **Ruby, React, React Native**, and **GraphQL** to build and maintain robust features at scale
- Championed **parallel updating feature**, improving **performance of bulk updates** with **6x speedup**
- **Responsible for all technical details** of the project, **balancing requirements** from different teams
- Rated as **top 5% of interns** by manager, highlighting **self-sufficiency and performance**

Co-founder, Swimhomes.ca  present
- Building a **modern, easy, automated**, experience for selling a home without Real Estate Agents
- Leveraged **Typescript, React, Stripe, Supabase** to create scalable service: Swimhomes.ca 
- Performed reachout to validate customer needs, secure partnerships, and explore opportunities

Full Stack Developer, Dawson Partners (FKA WhiteHorse Liquidity Partners) 05/2023 – 08/2024
- **Responsible for** all webscraping and data pipelines, from **architecture, interface, implementation**
- Built **dashboards** for end-users and monitoring using **React** and **Python Fastapi** backend
- Improved **webscraping** pipeline with **10x speedup** by **reverse-engineering** websites **APIs**
- **Pitched and implemented** system to eliminate duplicate data and **reduce processing times by 90%**
- Automated **CI/CD** to **build, test, and deploy** with pre-baked docker containers to **deploy faster**

Backend Developer, Dawson Partners (FKA WhiteHorse Liquidity Partners) 01/2023 – 05/2023
- Created **Python** webscraping pipeline with **Selenium, FastAPI**, and **NoSQL** on **Google Cloud**
- Designed **Python OOP** solution for **scraping** and data collection, **reducing costs by 90%**
- **Presented MVPs to leadership**, helping to secure funding to **grow team from solo to 5+**

Software Team Lead, University of Waterloo Alternate Fuels Team 09/2021 – 05/2024
- **Led team of 7** to develop adaptive cruise control and lane keeping for EcoCAR EV Challenge
- Created networked system using **Matlab, C++, Python**, using **CAN, ROS2** to connect systems
- Performed **sensor fusion** on LiDAR, RADAR, camera, and IMU data for increased accuracy

Machine Learning Research Assistant, University of Waterloo (Dr Mei Nagappan) 05/2022 – 01/2023
- Created a user-friendly chrome extension that uses **AI to detect fake news** with inhouse model
- Worked with **PyTorch** using **GPU acceleration** using slurm on Compute Canada **Supercomputers**
- **Scraped** website data to **create news datasets** for training and testing
- **Created pipeline** to clean data, **improving precision and recall** of the model by 20%
- **Collaborated with external researchers** to validate results and brainstorm new approaches

Projects

IMC Prosperity Algorithmic Trading Competition, Python
- Developed, backtested, and optimized trading strategies including arbitrage, and market making using mean reversion, options pricing, and position sizing models

FIX Parser & Order Matching Engine, C++ 
- Working through low level exchange operations to improve C++ and work through practical problems

Recommendation Model for Movies, Python, Pytorch, C++
- Created a custom **autoencoder Neural Network** with PyTorch, trained using **Local GPU** setup
- Added custom loss function for model, **increasing accuracy by 50%** compared to out-of-the-box solutions
- Built **performant multi-threading** program in **C++** to **5x data throughput** when pre-processing files for training NN

Raspberry Pi Keyboard Macro Bluetooth Relay, C, Bluetooth, Raspberry Pico W
- Created custom **embedded solution** to enable remapping keys for consistent keymappings across MacOS/Windows/Linux
- Developed man-in-the-middle device that intercepts, interprets macros, and relays keyboard events with minimal latency

ASA DataFest - Winner of Best Insight, Python, R
- **Analyzed data** using **Pandas** and **R** to find trends and offered insights on how to **increase efficacy by 20%**
- Gave a winning **presentation** with **insights** on pitfalls in the game and **offered actionable solutions** to improve

Jarvis Personal Assistant, Python, AI APIs, MCP, RAG
- Built a personal bot integrated with google drive with **RAG** to track schedules and tasks, deployed on Raspberry Pi